📄 Comparison of Two Current Mirror Configurations

🔹 Method 1: Symmetric Circuit (with R2 and V2)

🔧 Description

- Both branches are identical
- Each branch has:
  - Supply voltage = 12V
  - Resistor = 120Ω
  - Identical MOSFET
- Gates of both transistors are connected

📊 Observation

- Current through both branches is equal
- Current remains constant (~92 mA)

⚠️ Limitation

- Output voltage is **fixed**
- Circuit behaves as two identical parallel branches
- Does not test actual current mirror functionality

🔹 Method 2: Proper Current Mirror (with Vtest / DC Sweep)

🔧 Description

- Reference branch:
  - V1 = 12V
  - R1 = 10kΩ
  - M1 is diode-connected
- Output branch:
  - M2 drain connected to **test voltage (Vtest)**
  - Source grounded
  - Gate connected to M1
- DC sweep applied:

```
.dc V2 0 12 0.1
```

📊 **Observation**

- Reference current remains constant (~92 mA)
- Output current:
  - Varies at low voltage
  - Becomes constant after a certain voltage

✅ **Key Result**

- Current remains constant despite change in output voltage
- Confirms correct current mirror operation

🧠 **Important Concept**

Compliance Voltage: Minimum output voltage required for proper current mirroring.

Observation:

Look at **Id(M2)** (blue curve):

- Starts at 0 current at **0V**
- Rises sharply
- Becomes **flat (~1.1 mA)**

In our case compliance voltage is 0.2 to 0.3V as around it does the line become flat.

Summary:

The current mirror experiment was performed to verify current replication using MOSFETs. In the first configuration, both branches had identical resistors and supply voltages (R2 and V2), resulting in equal currents. However, this was due to symmetry and not actual current mirror operation, as the output voltage was fixed. In the second configuration, a proper current mirror was implemented by applying a variable test voltage at the output. A DC sweep showed that the output current becomes constant after a certain voltage, confirming correct mirroring behaviour. This minimum required voltage, known as the compliance voltage, was found to be approximately 0.2–0.3 V. Below this, the transistor operates in the triode region, and above it, in saturation, ensuring constant current.